

A Parity Theorem for Reverse-Complement-Invariant Minimum Decycling Sets in Binary de Bruijn Graphs

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Abstract

Let $B(2, k)$ be the directed binary de Bruijn graph on length- k words, and let reverse complement act by reversing a word and complementing every bit. A minimum decycling set is a smallest vertex set whose deletion makes the graph acyclic. We prove that a reverse-complement-invariant minimum decycling set exists if and only if k is even, for every $k \geq 2$.

The odd obstruction is a single primitive rotation cycle connecting two reverse-complement-fixed lower states in opposite directions. For even k , we begin with a tie-symmetrized Mykkeltveit selector. We prove directly that arbitrary choices on zero-embedding rotation cycles preserve acyclicity, choose those ties equivariantly, and establish the exact half-plane identity

$$\mathbf{1}_{\text{RC}(M)} - \mathbf{1}_M = \nabla \chi.$$

Here χ is the open-upper-half-plane indicator on lower words. A finite sequence of legal source firings removes this nonnegative symmetric defect two coordinates at a time while preserving minimum-decycling-set status, and terminates at a reverse-complement-fixed set. An independent exact implementation through $k = 12$ is reported only as convention-level assurance; it is not used in the all-order proof.

Contents

1	Introduction	2
2	Normalization and prior work	3
2.1	Words, graphs, and reverse complement	3
2.2	Pure cycling registers and the ordinary minimum	3
2.3	Firing convention	3
2.4	What is imported and what is proved here	4
3	The odd-order obstruction	4
4	The cyclotomic seed and zero-sector ties	5
4.1	The selector	5
4.2	Equivariant zero ties	6
5	The half-plane coboundary	6
6	Legal defect descent and even existence	7
7	The binary parity theorem	8
8	Exact validation and reproducibility	9

9	Boundaries and outlook	10
A	Convention dictionary	10
B	Technical lemmas on stable PCR	11
C	Finite validation boundaries	11

1 Introduction

Minimum decycling sets in directed de Bruijn graphs are both classical combinatorial objects and a source of finite-window sampling constructions. The ordinary minimum is governed by rotation cycles and was established by Mykkeltveit’s construction and later unavoidable-set formulations [4, 1]. Modern work has made the move structure of minimum decycling sets explicit [2]. Reverse-complement symmetry introduces a different constraint: a chosen word must be accompanied by the reverse complement of that word. This constraint is natural in canonicalized sequence processing and was stated directly in the k -nonical setting [3].

This paper settles the binary existence question.

Theorem 1.1 (Binary parity theorem). *For every binary word length $k \geq 2$, the directed de Bruijn graph $B(2, k)$ has a reverse-complement-invariant ordinary minimum decycling set if and only if k is even.*

The two directions have different mechanisms. At odd order, a primitive pure-rotation cycle has two reverse-complement-fixed lower endpoints. One deletion from that cycle cannot destroy both endpoint-to-endpoint directions, and reverse complement supplies the opposing residual path. At even order, we exploit a feature of the classical cyclotomic construction which is not visible from a generic symmetry average. After resolving the zero-embedding ties equivariantly, the reverse-complement defect of the Mykkeltveit set is the exact gradient of an open-half-plane indicator. Legal source firings then remove the defect monotonically.

The proof is constructive in the following precise sense. It gives an initial ordinary minimum set, an integral nonnegative defect potential, and a finite legal firing rule. It does not claim a new complexity bound or an optimal firing length.

Relation to prior terminology. The phrase “symmetric decycling set” has also been used for the ordinary Mykkeltveit construction based on opposite geometric half-planes [5]. We therefore use the unambiguous phrase *reverse-complement-invariant* throughout.

Evidence discipline. The all-order proof occupies Sections 3–7. Section 8 records an independent exact finite implementation only as a sign, convention, and mutation check. No finite census is a premise of Theorem 1.1.

Organization. Section 2 freezes the graph, rotation, and gradient conventions. Section 3 proves the odd obstruction. Sections 4–6 give the even construction. Section 7 composes the two halves and records exact scope. The appendices collect convention maps, technical details, and the finite assurance table.

Claim boundary. The theorem is binary on the even side. The odd obstruction holds for every finite alphabet with a fixed-point-free complement, but no alphabet-general even theorem, exact odd excess, biological performance statement, or priority claim is made here.

2 Normalization and prior work

2.1 Words, graphs, and reverse complement

Let $\mathcal{A}$ be a finite alphabet of even cardinality q , equipped with a fixed-point-free involution $a \mapsto \bar{a}$. The binary specialization is $\mathcal{A} = \{0, 1\}$ with $\bar{0} = 1$ and $\bar{1} = 0$. For a word $w = w_0 \cdots w_{k-1}$, define

$$\text{RC}(w)_i = \overline{w_{k-1-i}}.$$

The directed vertex de Bruijn graph $B(q, k)$ has vertex set $\mathcal{A}^k$ and an arc $u \rightarrow v$ exactly when $u_1 \cdots u_{k-1} = v_0 \cdots v_{k-2}$. Loops are retained. Reverse complement is a directed anti-automorphism:

$$u \rightarrow v \implies \text{RC}(v) \rightarrow \text{RC}(u).$$

A set $M \subseteq \mathcal{A}^k$ is a *decycling set* when $B(q, k) \setminus M$ is a directed acyclic graph (DAG), and is an *ordinary minimum decycling set* when its cardinality is minimum among all decycling sets. We abbreviate the latter to MDS. Reverse-complement invariance means $\text{RC}(M) = M$ as a literal set of labelled words.

The line-graph identity $B(q, k) = L(B(q, k-1))$ lets us regard each $w = w_0 \cdots w_{k-1}$ as the labelled lower-state arc

$$t(w) = w_0 \cdots w_{k-2} \longrightarrow h(w) = w_1 \cdots w_{k-1}.$$

Deleting M as vertices of $B(q, k)$ is equivalent to deleting those labelled arcs from $B(q, k-1)$; acyclicity is equivalent in the two representations.

2.2 Pure cycling registers and the ordinary minimum

Let $R(w_0 \cdots w_{k-1}) = w_1 \cdots w_{k-1} w_0$. A pure cycling register (PCR) is an orbit under R , using distinct rotations only. Its cardinality is the minimal period of its words. PCRs partition $\mathcal{A}^k$, and each is a directed cycle. Their number is the necklace count

$$N_q(k) = \frac{1}{k} \sum_{d|k} \varphi(d) q^{k/d}. \quad (1)$$

Every decycling set meets every PCR. The classical construction meets this lower bound, so the ordinary minimum cardinality is $N_q(k)$ [4, 1]. Consequently an MDS selects exactly one distinct word from each PCR. The converse is false: one-per-PCR is a local cardinality condition and can leave cycles composed of arcs from several PCRs. The binary homopolymers 0^k and 1^k are singleton PCR loops and are mandatory selections.

2.3 Firing convention

For a lower word $u \in \mathcal{A}^{k-1}$, write

$$\text{lc}(u) = \{au : a \in \mathcal{A}\}, \quad \text{rc}(u) = \{ua : a \in \mathcal{A}\}.$$

Here “lc” and “rc” mean left and right companion sets; the latter is not the reverse-complement operator RC. A forward firing, or F -move, is valid at u for a selected set M precisely when

$$\text{lc}(u) \subseteq M, \quad F_u(M) = (M \setminus \text{lc}(u)) \cup \text{rc}(u).$$

These are literal sets: any overlap between the companion sets, including the homopolymer-loop case, is retained rather than counted twice. The fact that a valid F -move preserves

ordinary-MDS status is the source-move result of Marçais–DeBlasio–Kingsford [2, Section 4.1, Proposition 2]. We import only this preservation fact, not any conjectural connectivity statement.

For an integer function q on lower states, our gradient sign is

$$(\nabla q)(w) = q(t(w)) - q(h(w)). \quad (2)$$

With this convention, a valid firing at u changes the selected-edge indicator by $\nabla \mathbf{1}_u$; the set-overlap convention makes this equality valid also at all homopolymer loops (the two loops in the binary specialization).

2.4 What is imported and what is proved here

We import the ordinary cardinality theorem and valid- F -move preservation. We prove the odd obstruction, arbitrary-zero-tie acyclicity of the chosen cyclotomic seed, equivariant tie selection, the exact coboundary identity, and the terminating defect descent. The 2024 k-nonical paper supplies a direct dated formulation of the symmetry-constrained problem [3]; it states that the minimum cardinality under the symmetry constraint was not known and uses integer linear programming for bounded orders. Our result settles the binary ordinary-necklace-bound existence question at every order. It does not replace that paper’s optimization, implementation, or window-guarantee analysis, and its bounded computations are not used as a theorem in our proof.

3 The odd-order obstruction

The negative direction is not specific to the binary alphabet.

Theorem 3.1 (Odd obstruction). *Let $\mathcal{A}$ be a finite alphabet with a fixed-point-free complement. For every odd $k = 2m + 1 \geq 3$, no reverse-complement-invariant decycling set in $B(q, k)$ has cardinality $N_q(k)$. Its cardinality exceeds $N_q(k)$ by at least two.*

Proof. Fix a complementary pair $a, \bar{a}$ and define two length- $2m$ lower states

$$f_m = (a\bar{a})^m, \quad g_m = (\bar{a}a)^m.$$

Both are fixed by reverse complement. Consider the length- k word

$$w_m = (a\bar{a})^m \bar{a}.$$

It contains m copies of a and $m + 1$ copies of $\bar{a}$. If it were a proper r -fold power, then $r > 1$ would divide both counts, contradicting $\gcd(m, m + 1) = 1$. Thus its PCR has full length k .

In the lower-state graph this PCR contains the direct arc $g_m \rightarrow f_m$ and, using all its other $k - 1$ arcs, a directed path from f_m to g_m . Any decycling set of cardinality $N_q(k)$ selects exactly one arc of this PCR. Deleting the direct arc leaves the long path; deleting any other arc leaves the direct path. Hence one of the two endpoint-to-endpoint directions remains.

Suppose the deletion set is reverse-complement invariant. Since both endpoints are fixed and reverse complement reverses arrows, the surviving path has a reverse-complement image directed from the other endpoint back to the first. Their concatenation is a nonempty closed directed walk, which contains a directed cycle. This contradicts decycling.

At odd length no word is reverse-complement fixed, so every invariant set has even cardinality. Reverse complement also acts freely on PCRs. Indeed, if an odd-period PCR

were setwise stable, then for some word w of odd minimal period d and some $s \in \mathbb{Z}/d\mathbb{Z}$ we would have $\text{RC}(w) = R^s w$. On rotation indices reverse complement induces the reflection $r \mapsto s - r$ modulo d . (The closely related position-index formula is $i \mapsto s - 1 - i$.) Every reflection of an odd cycle has a fixed index, which would force the corresponding letter of w to equal its own complement, contrary to the fixed-point-free hypothesis. Thus PCRs occur in free reverse-complement pairs, so $N_q(k)$ is even. Strict excess is therefore at least two. $\square$

The proof does not determine the exact odd excess. In particular, no formula inferred from small odd orders is needed anywhere in this paper.

4 The cyclotomic seed and zero-sector ties

For the remainder of the proof, let $k \geq 4$ be even and specialize to the binary alphabet. Put $\zeta = e^{2\pi i/k}$. For $w = x_0 \cdots x_{k-1}$ define

$$P(w) = \sum_{j=0}^{k-1} x_j \zeta^{j+1}, \quad W(w) = \sum_{j=0}^{k-1} x_j.$$

On a lower word $u = u_0 \cdots u_{k-2}$ define

$$\lambda(u) = \sum_{j=0}^{k-2} u_j \sin\left(\frac{2\pi(j+1)}{k}\right), \quad \chi(u) = \mathbf{1}[\lambda(u) > 0].$$

Lemma 4.1 (Covariance). *For every length- k word w ,*

$$P(Rw) = \zeta^{-1}P(w), \quad P(\text{RC } w) = -\overline{P(Rw)}, \quad P(R \text{RC } w) = -\overline{P(w)}.$$

Moreover

$$\text{Im } P(w) = \lambda(t(w)), \quad \text{Im } P(Rw) = \lambda(h(w)),$$

and $\lambda(\text{RC } u) = \lambda(u)$ on lower words.

Proof. The rotation identity follows by reindexing and using $\zeta^k = 1$. For reverse complement, the sum of all k th roots of unity is zero, so

$$P(\text{RC } w) = \sum_i (1 - x_{k-1-i}) \zeta^{i+1} = - \sum_j x_j \zeta^{-j} = -\overline{P(Rw)}.$$

Rotating once gives the third identity. The omitted last bit of $t(w)$ would multiply $\zeta^k = 1$ and contributes no imaginary part; the head identity follows by applying the same observation to Rw . Finally, reindexing the lower sum and using $\sin(2\pi(k-r)/k) = -\sin(2\pi r/k)$ proves lower-word invariance. $\square$

4.1 The selector

On each PCR with $P \neq 0$, select one word by the following rule:

1. select the point on the negative real axis, if it occurs;
2. otherwise select the unique w such that $\text{Im } P(w) < 0 < \text{Im } P(Rw)$.

On a PCR where P vanishes identically, select an arbitrary representative for now. Denote the resulting one-per-PCR set by M .

Lemma 4.2 (Full period and uniqueness). *Every nonzero embedded PCR has full period k , and the selector above is unique.*

Proof. If a word in a nonzero PCR had period $d < k$, then $P(R^d w) = \zeta^{-d}P(w) = P(w)$ would force $P(w) = 0$. The nonzero rotations therefore form the k vertices of a regular polygon. Since k is even, real vertices occur in antipodal pairs. If the negative ray is absent, there is no real vertex at all, and clockwise rotation gives exactly one strict crossing from the lower to the upper half-plane. Otherwise the negative-ray vertex is unique. $\square$

The zero sector must be handled explicitly: choosing its representatives carelessly could appear to affect cycles. The next theorem shows that it does not.

Theorem 4.3 (Arbitrary-zero-tie acyclicity). *For every choice of representatives on zero-embedding PCRs, the set M is an ordinary MDS.*

Proof. Define $Q(w) = P(w) - W(w)$. If $y = x_1 \cdots x_{k-1}a$ follows $w = x_0 \cdots x_{k-1}$ and $\delta = a - x_0$, then

$$P(y) = \zeta^{-1}P(w) + \delta, \quad W(y) = W(w) + \delta, \quad Q(y) = -W(w) + \zeta^{-1}P(w).$$

Thus $Q(y)$ is obtained by rotating $Q(w)$ clockwise by $2\pi/k$ about the real center $-W(w)$.

A residual path cannot move from $\text{Im } P \leq 0$ to $\text{Im } P > 0$: a strict lower-to-upper crossing is selected, and a departure from the negative ray is selected. From the positive ray the clockwise rotation enters the lower half-plane; from the origin the next imaginary coordinate is zero. A residual directed cycle must therefore lie entirely in the open upper half-plane or in the closed lower half-plane.

Write $\theta = 2\pi/k$ and $P(w) = \rho e^{i\phi} \neq 0$. Along a residual edge within the upper region,

$$\text{Re } Q(y) - \text{Re } Q(w) = 2\rho \sin(\theta/2) \sin(\phi - \theta/2) > 0.$$

Along a residual edge within the closed lower region the same quantity is strictly negative: the negative-ray starting point was selected, and the remaining angular interval is $\pi + \theta \leq \phi \leq 2\pi$. Hence neither region contains a residual cycle unless every edge of that cycle begins at $P(w) = 0$.

If consecutive labels w and y both have zero embedding, then $P(y) = a - x_0$ forces $a = x_0$. Every step is a pure rotation. Such a cycle is exactly a zero-embedding PCR, from which one word was selected. This is again impossible in the residual. Therefore the residual lower graph is a DAG. Since M selects one word from every PCR, it attains the lower bound $N_2(k)$ and is an ordinary MDS. $\square$

4.2 Equivariant zero ties

Lemma 4.1 shows that reverse complement permutes the zero-embedding PCRs. On a paired orbit $\{C, \text{RC}(C)\}$, choose $w \in C$ and $\text{RC}(w)$ in the mate. On an RC-stable PCR, choose either of its two RC-fixed rotations; Appendix B proves that exactly two exist. These choices make the zero-sector part of M invariant without altering the MDS property from Theorem 4.3.

5 The half-plane coboundary

Let $m = \mathbf{1}_M$ for the tie-symmetrized selector of Section 4. We use

$$\mathbf{1}_{\text{RC}(M)}(w) := m(\text{RC}(w)),$$

which agrees with set-image notation because reverse complement is an involution.

Theorem 5.1 (Exact reverse-complement defect). *For every length- k word w ,*

$$m(\text{RC } w) - m(w) = \chi(t(w)) - \chi(h(w)) = (\nabla \chi)(w). \quad (3)$$

Equivalently, $\mathbf{1}_{\text{RC}(M)} - \mathbf{1}_M = \nabla \chi$.

Proof. Set

$$A = \text{Im } P(w), \quad B = \text{Im } P(Rw).$$

Lemma 4.1 sends the ordered imaginary coordinates of $\text{RC}(w)$ to (B, A) . If A and B are nonzero, the selector rule gives

$$m(\text{RC } w) - m(w) = \mathbf{1}[B < 0 < A] - \mathbf{1}[A < 0 < B] = \mathbf{1}[A > 0] - \mathbf{1}[B > 0].$$

It remains to check the rays. If $A = 0$, write $P(w) = r \in \mathbb{R}$. Then $B = -r \sin(2\pi/k)$. For $r < 0$ both sides are -1 , while for $r > 0$ both are zero. If $B = 0$, write $P(Rw) = r \in \mathbb{R}$. For $r > 0$ both sides are $+1$, while for $r < 0$ both are zero. When $k > 2$, $A = B = 0$ with $P(w) \neq 0$ is impossible because $P(Rw) = \zeta^{-1}P(w)$ and ζ is not real. In the zero sector, the chosen ties are RC invariant, so the left side vanishes; both lower imaginary coordinates vanish as well.

Finally Lemma 4.1 identifies the last sign difference with $\chi(t(w)) - \chi(h(w))$, with the gradient convention (2). $\square$

Equation (3) is the bridge between the cyclotomic seed and an exact legal move path. Its right-hand side is not a heuristic score: it is the literal indicator defect, edge by edge. The use of an *open* upper half-plane and the chosen negative-ray convention are load-bearing.

6 Legal defect descent and even existence

We now convert the exact gradient defect into a finite sequence of valid F -moves. Reverse complement carries an ordinary MDS to an ordinary MDS, because it reverses directed cycles bijectively.

Lemma 6.1 (Positive-potential firing). *Let A and N be ordinary MDSs and suppose*

$$\mathbf{1}_N - \mathbf{1}_A = \nabla q$$

for an integral lower-state potential $q \geq 0$ with minimum zero. If $A \neq N$, some lower state u with $q(u) > 0$ admits a valid F -move from A .

Proof. Assume otherwise. At every positive state u , failure of the source condition supplies an incoming nonloop residual arc $v \rightarrow u$; a residual loop is already impossible because A is decycling. Since that arc is not in A ,

$$\mathbf{1}_N(v \rightarrow u) = q(v) - q(u) \in \{0, 1\},$$

so $q(v) \geq q(u) > 0$. Repeating the predecessor choice in the finite set of positive states eventually creates a directed cycle entirely outside A , contradicting that A is a decycling set. $\square$

Lemma 6.2 (Defect-paired update). *Suppose the current ordinary MDS A satisfies*

$$\mathbf{1}_{\text{RC}(A)} - \mathbf{1}_A = \nabla q, \quad q \geq 0, \quad q \circ \text{RC} = q,$$

with minimum zero. If A is not RC invariant, let u be an enabled positive state from Lemma 6.1, and fire u . The new ordinary MDS A' satisfies

$$\mathbf{1}_{\text{RC}(A')} - \mathbf{1}_{A'} = \nabla(q - \mathbf{1}_u - \mathbf{1}_{\text{RC}(u)}). \quad (4)$$

The new potential is nonnegative, RC invariant, has minimum zero, and its coordinate sum is two smaller.

Proof. A valid firing changes $\mathbf{1}_A$ by $\nabla \mathbf{1}_u$ and preserves ordinary-MDS status. Since reverse complement reverses lower arcs,

$$\text{RC}(\nabla \mathbf{1}_u) = -\nabla \mathbf{1}_{\text{RC}(u)}.$$

Subtracting the update of A from the update of its RC image gives (4). Positivity is paired because $q \circ \text{RC} = q$. The lower words have odd length $k - 1$, so none is RC fixed; u and $\text{RC}(u)$ are distinct positive states. Subtracting one at those two coordinates preserves nonnegativity and invariance. A previously zero coordinate is untouched, so the minimum remains zero. $\square$

Theorem 6.3 (Even existence). *For every even $k \geq 4$, $B(2, k)$ has an RC-invariant ordinary MDS.*

Proof. Start from the tie-symmetrized ordinary MDS M constructed in Section 4. Theorem 5.1 supplies the initial potential $q = \chi$, which is binary, nonnegative, RC invariant, and has minimum zero.

If the current set is not RC invariant, apply Lemmas 6.1 and 6.2. Every step preserves ordinary-MDS status and lowers the nonnegative integer $\sum_u q(u)$ by two. Hence the process terminates. At termination the set must be RC invariant: otherwise another positive firing exists. Equivalently, if the terminal defect were zero while q were nonzero, then $\nabla q = 0$; connectedness of the lower binary de Bruijn graph would make q constant, and its retained minimum zero would force $q = 0$. Thus the terminal ordinary MDS is literally fixed by reverse complement. $\square$

Only the selected state u is fired at each step. The word “paired” in Lemma 6.2 refers to the forced two-coordinate update of the RC defect potential, not to an unproved simultaneous-move operation.

For clarity, the constructive content of the proof can be read as the following exact procedure.

1. Construct the Mykkeltveit seed and resolve its zero PCRs by the RC-equivariant tie rule of Section 4.
2. Compute the binary potential $q = \chi$ from the open half-plane.
3. While $q \neq 0$, choose any positive lower state enabled by Lemma 6.1, fire that single state, and replace q by $q - \mathbf{1}_u - \mathbf{1}_{\text{RC}(u)}$.
4. Return the terminal selected set.

The proof establishes existence of a choice at every iteration and exact termination. This description is not a claim of an optimized implementation or of a particular complexity bound.

7 The binary parity theorem

The even base case is immediate. At $k = 2$ the set

$$M_2 = \{00, 01, 11\}$$

selects one word from each of the three PCR, is RC invariant, and leaves only the vertex 10, with no residual loop. It is therefore an ordinary MDS.

Proof of Theorem 1.1. For even $k \geq 4$, Theorem 6.3 supplies an RC-invariant ordinary MDS; the preceding paragraph covers $k = 2$. For odd $k \geq 3$, Theorem 3.1, specialized to the binary complement, rules out an RC-invariant decycling set at the ordinary minimum. These cases exhaust the stated domain. $\square$

The logical dependencies are worth making explicit:

$$\begin{array}{c}
\text{ordinary PCR lower bound and MDS construction} \quad + \quad \text{valid } F\text{-move preservation} \\
\Downarrow \\
\text{arbitrary-zero-tie seed} \implies \text{exact half-plane coboundary} \implies \text{finite legal defect descent} \\
\Downarrow \\
\text{all even binary orders, combined independently with the odd bridge.}
\end{array}$$

No component-count conjecture, solver outcome, bounded-order pattern, or connectivity claim enters this chain.

It is useful to name the quantity separated by the theorem. Let $\tau_{\text{RC}}(k)$ denote the minimum cardinality of an RC-invariant decycling set in $B(2, k)$ and define the *symmetry premium*

$$\Delta_2(k) = \tau_{\text{RC}}(k) - N_2(k).$$

The theorem says exactly that $\Delta_2(k) = 0$ for even k , whereas Theorem 3.1 gives $\Delta_2(k) \geq 2$ for odd k . No exact formula for the odd premium is asserted.

8 Exact validation and reproducibility

An independent finite validation was used to audit the convention map, not to infer the theorem. Two implementations reconstructed the selector, zero ties, residual DAG, edgewise coboundary identity, firing path, and terminal RC-fixed MDS with exact cyclotomic arithmetic at $k = 2, 4, 6, 8, 10, 12$. One implementation generated coordinates from an integer recurrence in $\mathbb{Z}[2 \cos(2\pi/k)]$; the other used independent frozen coordinate tables and a separate recursive DFS DAG oracle.

Table 1: Exact finite convention-level assurance. “F moves” is the number of legal firings in the deterministic validation trajectory.

k	PCRs	zero PCRs	stable zero	paired zero	$ \text{supp } \chi $	F moves
2	3	0	0	0	0	0
4	6	3	1	1	2	1
6	14	5	1	2	10	5
8	36	6	2	2	52	26
10	108	9	1	4	220	110
12	352	19	5	7	940	470

Across the nontrivial rows, the check contains 42 zero PCRs, 1,224 positive lower states, and 612 legal firings. Seven scientific mutations were actually applied and rejected: right rotation, reversed gradient sign, closed half-plane, positive-axis convention, broken zero-sector pairing, unpaired potential update, and reversed firing. Seven additional envelope mutations were also rejected. The stable scientific payload digest is

3bbec32e9d37706e98db9bb61192d12653a6adb8f419a2f1e816b31cd664d9b5.

This finite panel checks that an implementation agrees with the printed signs, rays, and tie conventions. The quantifiers in Theorem 1.1 are discharged only by the written proof.

9 Boundaries and outlook

The result isolates a sharp parity effect for literal reverse-complement invariance at the ordinary binary necklace bound. Several nearby questions remain outside its scope.

- The odd proof gives $\Delta_2(k) \geq 2$ but no exact formula for the symmetry premium.
- The even construction is binary. Although the odd obstruction is alphabet-general, an even construction for larger fixed-point-free complement alphabets would require a separate argument.
- The firing proof supplies termination but no optimized step count or algorithmic complexity theorem.
- No biological, empirical sketching-performance, or clinical claim follows from this existence theorem.

The direct predecessor literature establishes ordinary MDS constructions, move structure, and the reverse-complement-constrained design problem [4, 1, 2, 3]. We are not aware of a direct later proof of the same binary parity statement. Our targeted primary-source search was not logged at exhaustive-query level; it therefore supplies neither a priority claim nor a claim of literature exhaustion. Any future novelty assessment must remain within that documented search boundary.

One conceptual lesson of the proof is that exact equivariance need not be present in the classical seed itself. It can instead appear as an integral coboundary with a nonnegative, symmetry-invariant potential, provided the ambient move system realizes every positive defect by a legal descent. The odd bridge shows why no analogous minimum construction can cross the parity boundary.

A Convention dictionary

Object	Frozen convention
de Bruijn orientation	$u \rightarrow v$ when the suffix of u equals the prefix of v
rotation	$R(w_0 \cdots w_{k-1}) = w_1 \cdots w_{k-1} w_0$
reverse complement	reverse order, then complement; a directed anti-automorphism
PCR	set of distinct rotations; never an unfolded length- k list
lower-state arc	$w : t(w) = w_0 \cdots w_{k-2} \rightarrow h(w) = w_1 \cdots w_{k-1}$
gradient	$(\nabla q)(w) = q(t(w)) - q(h(w))$
half-plane	open upper half-plane; negative-real-axis selector
set equality	literal labelled-word equality unless stated otherwise
MDS	ordinary cardinality minimum plus a literal residual DAG
finite evidence	exact at its frozen orders; never an all-order premise

The paper excludes $k = 1$. At that order the lower-state edge presentation degenerates and is not part of the frozen normalization.

B Technical lemmas on stable PCRs

Lemma B.1. *Let C be an RC-stable binary PCR at even length k , and let d be its minimal period. Then d is even and C contains exactly two RC-fixed words.*

Proof. Choose $w \in C$ and $s \pmod{d}$ with $\text{RC}(w) = R^s w$. Stability pairs zeros and ones in the period, so d is even. In cyclic position notation the identity reads

$$w_r = \overline{w_{s-1-r}}.$$

The positional involution $r \mapsto s-1-r$ cannot have a fixed point, because that would force a bit to equal its complement. Since d is even, the congruence $2r \equiv s-1 \pmod{d}$ has a solution exactly when s is odd; hence s is even.

A rotation $R^i w$ is RC fixed precisely when

$$\text{RC}(R^i w) = R^{-i} \text{RC}(w) = R^{s-i} w = R^i w,$$

or $2i \equiv s \pmod{d}$. With d and s even, this congruence has exactly $\gcd(2, d) = 2$ solutions modulo d . $\square$

Lemma B.2. *On lower binary words of odd length, reverse complement has no fixed point.*

Proof. The central bit of a fixed odd-length word would have to equal its own complement. $\square$

Remark B.3. These local facts guarantee equivariant choices on stable zero PCRs. They do not by themselves imply that an arbitrary RC-compatible one-per-PCR transversal is decycling; the global DAG argument remains essential.

C Finite validation boundaries

The finite validation summarized in Table 1 has four epistemic roles:

1. verify the left-rotation and gradient conventions;
2. exercise negative-ray, strict-crossing, and zero-sector cases;
3. replay every legal firing and terminal invariant in the declared finite panel;
4. reject mutations that would otherwise look like harmless convention changes.

It does not establish a recurrence, interpolate in k , or replace any displayed lemma. Runtime, file-envelope, and platform hashes are deliberately kept separate from the stable scientific payload digest.

References

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